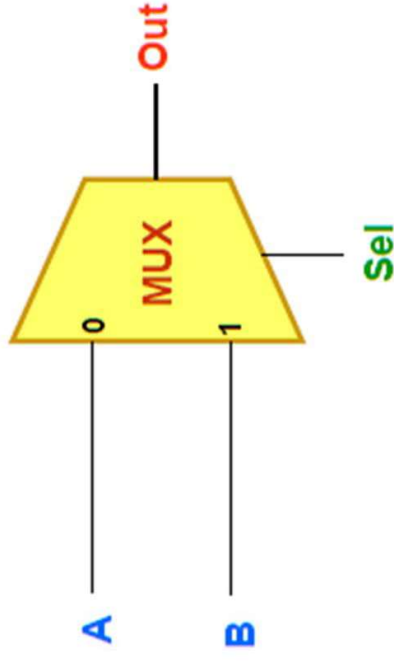


Design Representation (Example 1)



- **Multiplexer:** Choose one of two inputs based on a control input
- **Sel:** Select line (it is a control input)
- **A,B:** Data Inputs
- **Out:** Multiplexer Output



Design Representation (Example 1)



Truth Table

Sel	A	B	Out
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

- If **Sel** is '0', choose **A** and pass it on to **Out**
- If **Sel** is '1', choose **B** and pass it on to **Out**
- The value of **A** and **B** does not matter
 - Pass both 0's and 1's

Characteristic Table

Sel	Out
0	A
1	B

Boolean Equation

$$Out = \overline{Sel} \cdot A + Sel \cdot B$$

Gate-level Modeling



- A circuit is specified by its logic gates and their interconnection.
- Verilog recognizes 12 basic gates as predefined primitives.
- The logic values of each gate may be
 - **0** : zero, logic low
 - **1** : one, logic high
 - **X** : unknown
 - **Z** : high impedance

Verilog – Gate level Description



```
module simple_circuit(A,B,C,x,y);
```

```
input A,B,C;
```

```
output x,y;
```

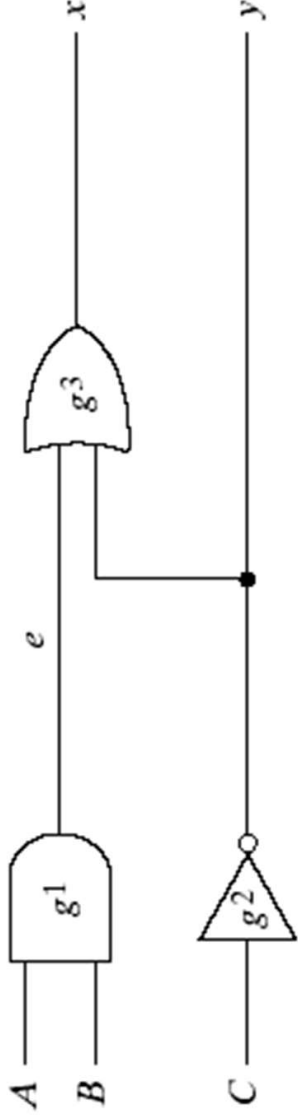
```
wire e;
```

```
and g1 (e,A,B);
```

```
not g2 (y,C);
```

```
or g3 (x,e,y);
```

```
endmodule
```



Verilog – Gate level Description



```
module multiplexer (A, B, Sel, Out);
```

```
input A, B, Sel;
```

```
output Out;
```

```
wire Selbar, tmp1, tmp0;
```

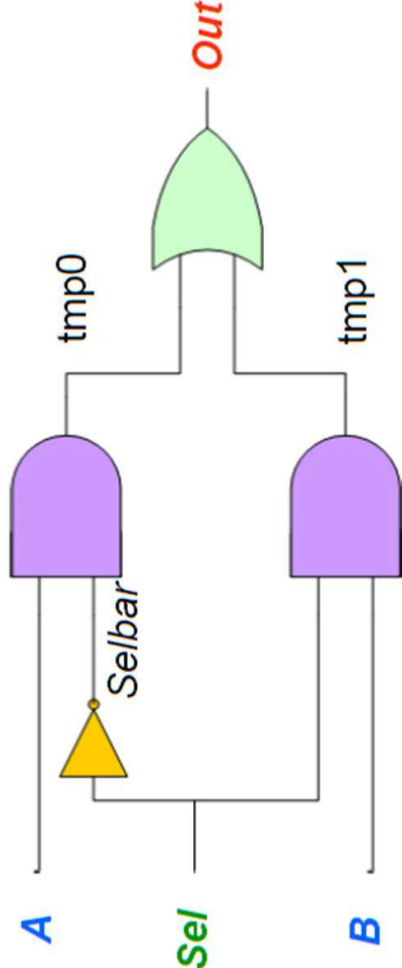
```
and a0 (tmp1, B, Sel);
```

```
not (Selbar, Sel);
```

```
and a1 (tmp0, A, Selbar);
```

```
or (Out, tmp0, tmp1);
```

```
endmodule
```



Verilog – Gate Delays



- Sometimes it is necessary to specify the amount of delay from the input to the output of gates.
- In Verilog, the delay is specified in terms of time units and the symbol #.
- The association of a time unit with physical time is made using ***timescale*** compiler directive.
- Compiler directive starts with the “backquote (`)” symbol.

**``timescale`** `1ns/100ps`

- The first number specifies the *unit of measurement* for time delays.
- The second number specifies the *precision* for which the delays are rounded off, in this case to 0.1ns.

Verilog – Gate level Description with delay



```
module circuit_with_delay (A,B,C,x,y);  
  input  A,B,C;  
  output x,y;  
  wire   e;  
  and #30 g1(e,A,B);  
  or  #20 g3(x,e,y);  
  not #10 g2(y,C);  
endmodule
```

